

1 Chapter 1

Nick's Question 1.1 Proof of lemma 1.8.4

The proof is very nice.

2 Chapter 2

Nick's Question 2.1 In the proof of Theorem 2.4.3 on the bottom of page 34 there are some claims made that I am unable to verify why they are true. First, we are considering

$$X_t = X_0 + \int_0^t u(s)dB_s$$

and here $u(s)$ only satisfies

$$\mathbb{P}\left(\int_0^t u(s)^2 ds < \infty\right) = 1$$

so, as in the Kuo text, this stochastic integral exists and a limit in probability of a sequence of stochastic integrals $\int_0^t u_n(s)dB_s$ where the u_n are simple and are in L^2_{ad} and hence $\int_0^t u_n(s)dB_s$ is a martingale and we know the first and second moment. But for the limit $\int_0^t u(s)dB_s$ we know nothing about the moments. But in this text, it is claimed that

$$\mathbb{E}\left[\int_{t_i}^{t_{i+1}} u^2(s)ds - \left(\int_{t_i}^{t_{i+1}} u(s)dB_s\right)^2 \middle| \mathcal{F}_i\right] = 0$$

and I have never seen this claimed in any other text.

Note that u is bounded (we are assuming localization has been done). Hence, all moments exists.

Nick's Question 2.2 In the proof of theorem 2.5.1, why is it true that

$$E\left(\int_0^t |u'_n(B_s) - 1_{(x,\infty)}(B_s)|^2 ds\right) \leq \int_0^t \mathbb{P}\left(|B_s - x| \leq \frac{2}{n}\right) ds$$

Please explore the mollifier f and f_n . Plot these u'_n and see how could it differ from $1_{(x,\infty)}$.

Nick's Question 2.3 What is the motivation on page 41 behind the following definition of the backward ito integral

$$\int_0^T u(s)\hat{d}B_s = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-2} \left(\frac{n}{T} \int_{(j+1)T/n}^{(j+2)T/n} u(s)ds\right) (B_{(j+1)T/n} - B_{jT/n})$$

The integral in the parenthesis is the average of u_s on $((j+1)T/n, (j+2)T/n)$.

Let's skip this part. I know nothing on the backward SDE and not particular interested either. It has applications in finance. If you don't do finance, I don't see the point here.

Nick's Question 2.4 (Problem 2.7)

This problem follows easily from Levy's Characterization of Brownian Motion. But I can't figure out how to do this problem with the hint given.

This is a nice problem. The hints is to ask you to find the distribution of $X_t - X_s$ through the characteristic function. Once apply Ito's formula, take expectation and remove the martingale part, you should arrive at an integral equation. Take derivatives, you arrive at a differential equation. The solution for $a_s(t) := \mathbb{E}(e^{i\lambda(X_t - X_s)})$ is

$$a_s(t) = e^{-\frac{\lambda^2}{2}(t-s)}.$$

Nick's Question 2.5 (Problem 2.15)

First we consider the 2 - d Ito process

$$\begin{pmatrix} M_s \\ N_s \end{pmatrix} = \begin{pmatrix} E(X^p) \\ E(Y^{q'}) \end{pmatrix} + \int_0^t \begin{pmatrix} \varphi_s & 0 \\ 0 & \psi_s \end{pmatrix} \begin{pmatrix} d\hat{B}_s \\ dB_s \end{pmatrix}$$

and then following the hint we apply Ito's formula to $f(x, y) = x^{1/p}y^{1/q'}$. We get the following derivatives:

- $f_x = \frac{1}{p}x^{\frac{1}{p}-1}y^{\frac{1}{q'}}$
- $f_{xx} = \frac{1}{p}\left(\frac{1}{p}-1\right)x^{\frac{1}{p}-2}y^{\frac{1}{q'}}$
- $f_y = \frac{1}{q'}x^{\frac{1}{p}}y^{\frac{1}{q'}-1}$
- $f_{yy} = \frac{1}{q'}\left(\frac{1}{q'}-1\right)x^{\frac{1}{p}}y^{\frac{1}{q'}-2}$

Now putting this into Ito's Formula gives us

$$\begin{aligned} M_t^{1/p} N_t^{1/q'} &= \|X\|_p \|Y\|_{q'} + \frac{1}{p} \int_0^t M_s^{\frac{1}{p}-1} N_s^{\frac{1}{q'}} \varphi_s d\hat{B}_s + \frac{1}{q'} \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-1} \psi_s dB_s \\ &\quad + \frac{1}{2} \frac{1}{p} \left(\frac{1}{p}-1\right) \int_0^t M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}} \varphi_s^2 ds + \frac{1}{2} \frac{1}{q'} \left(\frac{1}{q'}-1\right) \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-2} \psi_s^2 ds \end{aligned}$$

I am not really sure where to continue from here. I see that if we let $t \rightarrow \infty$ then $M_t^{1/p} N_t^{1/q'} \rightarrow XY$ so if we took the expected value of both sides then we would partially have what we are looking for since $E(XY)$ is on the left and $\|X\|_p \|Y\|_{q'}$ is on the right. But I am not sure about the other terms. Im guessing the Ito integrals in the right have 0 expectation but I am not sure if that is in fact true since I dont see how it is easily seen that the integrands are in $L^2(\mathcal{P})$.

What I did next was see what $\hat{B}_s$ was equal too and for this we define another Ito Process

$$\begin{pmatrix} B_s \\ \tilde{B}_s \end{pmatrix} = \int_0^t \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} dB_s \\ d\tilde{B}_s \end{pmatrix}$$

and applied Ito's formula to $f(t, x, y) = e^{-t}x + \sqrt{1 - e^{-2t}}y$ which gives us

$$\hat{B}_s = e^{-t}B_s + \sqrt{1 - e^{-2t}}\tilde{B}_s = \int_0^t \frac{e^{-2s}}{\sqrt{1 - e^{-2s}}}\tilde{B}_s - e^{-s}B_s ds + \int_0^t e^{-s}dB_s + \int_0^t \sqrt{1 - e^{-2s}}d\tilde{B}_s$$

However I am not sure if this step is necessary.

The followings are some of my thoughts: First of all, we need the absolute values:

$$\begin{pmatrix} M_s \\ N_s \end{pmatrix} = \begin{pmatrix} \mathbb{E}(|X|^p) \\ \mathbb{E}(|Y|^{q'}) \end{pmatrix} + \int_0^t \begin{pmatrix} \varphi_s & 0 \\ 0 & \psi_s \end{pmatrix} \begin{pmatrix} d\hat{B}_s \\ dB_s \end{pmatrix}.$$

Second, you forget

$$f_{xy} = \frac{1}{p} \frac{1}{q'} x^{-1/(p')} y^{-1/q'}.$$

Hence, by Ito's formula, we will have

$$\begin{aligned} M_t^{1/p} N_t^{1/q'} &= \|X\|_p \|Y\|_{q'} + \frac{1}{p} \int_0^t M_s^{\frac{1}{p}-1} N_s^{\frac{1}{q'}} \varphi_s d\hat{B}_s + \frac{1}{q'} \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-1} \psi_s dB_s \\ &\quad + \frac{1}{2} \frac{1}{p} \left(\frac{1}{p} - 1 \right) \int_0^t M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}} \varphi_s^2 ds + \frac{1}{2} \frac{1}{q'} \left(\frac{1}{q'} - 1 \right) \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-2} \psi_s^2 ds \\ &\quad + \frac{1}{pq'} \int_0^t M_s^{-\frac{1}{p'}} N_s^{-\frac{1}{q}} \psi_s \phi_s d\langle B, \hat{B} \rangle_s \\ &=: \|X\|_p \|Y\|_{q'} + \sum_{i=1}^5 I_i. \end{aligned}$$

Upon taking expectation on both sides, we see that

$$LHS = \mathbb{E}(|X_t Y_t|) \geq \mathbb{E}(X_t Y_t).$$

We only need to prove that

$$\mathbb{E} \left(\sum_{i=1}^5 I_i \right) \leq 0,$$

which is clear for $i = 1, 2$, in which case we have the expectation equal to zero. Now for the rest three parts, we see that

$$I_3 + I_4 + I_5 = -\frac{1}{2pq'} M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}-2} \left(N_s^2 \varphi_s^2 \frac{q'}{p'} - 2M_s N_s \phi_s \psi_s e^{-s} + M_s \psi_s^2 \frac{p}{q} \right),$$

where we have used the fact that $d\langle B, \hat{B} \rangle_s = e^{-s} ds$. Hence, the condition become the condition that the quadratic form has no solution:

$$\Delta = 4e^{-2s} - 4\frac{q'p}{p'q} = 4 \left(e^{-2s} - \frac{p-1}{q-1} \right) \leq 0$$

which is equivalent to

$$q \leq e^{2s}(p-1) + 1.$$

Hence, the largest q that makes the inequality work is $q = e^{2s}(p-1) + 1$.

3 Chapter 3

3.1 Reading Material

Nick's Question 3.1 (Smooth Cylindrical Random Variable) Following chapter 3 of the Kuo text, denote C the Wiener space of continuous functions from $[0,1]$ to $\mathbb{R}$, $\mathcal{R}$ the collection cylindrical sets

$$A = \{\omega \in C : (\omega(t_1), \dots, \omega(t_n)) \in U \mid t_i \in [0, 1], n \in \mathbb{N}, U \in \mathcal{B}(\mathbb{R}^n)\}$$

and μ the associated Wiener measure. Consider the coordinate processes $B(t, \omega) = \omega(t)$ which are Brownian motions under $(C, \mathcal{B}(\mathcal{R}), \mu)$. There is a theorem that can be found [here](#).

Theorem : The random variables

$$\{f(B(t_1, \cdot), \dots, B(t_n, \cdot)) \mid n \in \mathbb{N}, f \in \mathcal{S}(\mathbb{R}^n)\}$$

are dense in $L^2(C, \mu)$ where $\mathcal{S}(\mathbb{R}^n)$ are the Schwartz functions.

How does this relate to $\mathcal{S}$ defined on page 51 of the intro to Malliavin textbook? How are the cylindrical random variables defined in this case? Why is $\mathcal{S}$ dense in $L^2(\Omega)$? Even in Diffusions Markov Processes and Martingales - Roger and Williams, they define the cylindrical sigma field in the same way Kuo does- through the use of coordinate processes.

Nick Solution.

density

Theorem 3.1. Consider the probability space $(\Omega, \mathcal{F}, P)$ where $\mathcal{F}$ is the standard Brownian filtration. Then $\{f(B(h_1), \dots, f(B(h_n)), n \in \mathbb{N}, f \in C_p^\infty(\mathbb{R}^n), h_i \in H := L^2(\mathbb{R}^+)\}$ is dense in $L^2(\Omega)$.

Proof: This proof follows the same scheme as the solution found [here](#) which utilizes Dynkin's multiplicative system theorem. First, we can easily show that the sigma field generated by $\{B(h)\}_{h \in H}$ generates $\mathcal{F}$. This is true since $1_{[0,t]} \in H$ and $B(1_{[0,t]}) = B_t$. Second, since $\mathcal{S}(\mathbb{R}) \subset C_p^\infty(\mathbb{R})$ and since for every $h \in H$ there exists functions $f_n \in \mathcal{S}(\mathbb{R})$ such that $f_n \rightarrow h$ in $L^2(\mathbb{R})$ then by the Wiener Isometry we have $E|\int_0^\infty h(s) - f_n(s)dB_s|^2 = \int_0^\infty |h(s) - f_n(s)|^2 ds \rightarrow 0$. This implies that the sigma field generated by $\{f(B(h))\}$ where $h \in H$ and $f \in C_p^\infty(\mathbb{R})$ is the same as $\mathcal{F}$.

Now we let $M = \{f(B(h_1), \dots, B(h_n))\}$. We have already shown that $\sigma(M)$ is the same as $\mathcal{F}$. Also, M is closed under pointwise multiplication. Let $E := \bar{M}$ where is closure is with respect to the $L^2(\Omega)$ norm. Let $H := E \cap L^\infty(\Omega)$. We have $M \subset H$. It can be shown that H is closed with respect to bounded pointwise convergence and also that H contains all constant functions. Thus, the hypothesis is Dynkin's multiplicative theorem has been satisfied and this implies that H contains all $\sigma(M) = \mathcal{F}$ measurable functions. But this means that is contains all bounded Borel measurable functions. But by the definition of H , this implies that E contains all the bounded borel measurable functions. We know that the all $X \in L^2(\Omega)$ can be approximated in the $L^2(\Omega)$ norm by bounded measureable functions. Also since E is closed and contains the bounded measureable functions, then E itself must be equal to $L^2(\Omega)$. Therefore we conclude that $\bar{M} = E = L^2(\Omega)$. □

Nick's Question 3.2 Why is $(D_t F)_{t \geq 0}$ a stochastic processes in $L^2(\Omega; H)$ where $H = L^2(\mathbb{R}^+)$? $D_t F$ is defined as

$$D_t F = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(B(h_1), \dots, B(h_n)) h_i(t)$$

where $h_i \in H$ and $B(h_i) = \int_0^\infty h_i dB_s$ is the standard Wiener Integral. So isnt $D_t F$ real

valued since we are evaluating the $h_i(\cdot)$ at t ? Then shouldn't the $(D_t F)_{t \geq 0}$ be a $L^2(\Omega; \mathbb{R})$ valued stochastic process?

I believe that this is just a typo in the text

Nick's Question 3.3 (Proposition 3.2.4)(All questions have been solved with work below)

1. Why does $D_h(\delta(u)) = \langle u, h \rangle + \delta(D_h u)$? (I have answered this below)
2. Also, at the bottom of page 52, why is it that (I have answered this below)

$$E \left(\sum_{i,j=1}^{\infty} D_{e_i} \langle u, e_j \rangle D_{e_j} \langle v, e_i \rangle \right) = E \left(\int_0^{\infty} \int_0^{\infty} D_s u_t D_t v_s ds dt \right)$$

3. In the proof of property 3.3 why do we have (Answered below)

$$E \left(\sum_{i=1}^{\infty} \langle v, e_i \rangle \left(\langle u, e_i \rangle + \delta(D_{e_i}(u)) \right) \right) = E(\langle u, v \rangle) + E \left(\sum_{i,j=1}^{\infty} D_{e_i} \langle e, e_j \rangle D_{e_j} \langle v, e_i \rangle \right)$$

Proof of $D_h(\delta(u)) = \langle u, h \rangle + \delta(D_h u)$.

Since $u = \sum F_j h_j$ we just apply the definition of δ to get

$$\begin{aligned} D_h(\delta(u)) &= D_h \left(\sum_{j=1}^n F_j B(h_j) - \sum_{j=1}^n \langle DF_j, h_j \rangle \right) \\ &= \sum_{j=1}^n D_h(F_j B(h_j)) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \\ &= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \end{aligned} \quad (3.1) \quad \boxed{\text{P:3.2.4-1}}$$

where the third inequality follows from the product rule and $D_h(B(h_j)) = \langle h, h_j \rangle$. Now we focus on the last term from the last line above (3.1).

$$\sum_{j=1}^n D_h \langle DF_j, h_j \rangle = \sum_{j=1}^n \langle D_t \langle DF_j, h_j \rangle, h(t) \rangle \quad (3.2) \quad \boxed{\text{P:3.2.4-6}}$$

Before we continue, we simplify $D_t \langle DF_j, h_j \rangle$. We may assume that $F_j = f^j(B(h_1^j(t)), \dots, B(h_n^j(t)))$ where $f^j \in C_p^\infty(\mathbb{R}^n)$ and $h_i^j \in H$.

$$\begin{aligned} D_t \langle DF_j, h_j \rangle &= D_t \left\langle \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(s), h_j(s) \right\rangle \\ &= \sum_{i=1}^n \langle h_i^j(s), h_j(s) \rangle D_t \left(\frac{\partial f^j}{\partial x_i} \right) \end{aligned} \quad (3.3) \quad \boxed{\text{P:3.2.4-2}}$$

We now plug (3.3) into (3.2) above to get

$$\sum_{j=1}^n \langle D_t \langle DF_j, h_j \rangle, h(t) \rangle = \sum_{j=1}^n \sum_{i=1}^n \langle h_i^j(s), h_j(s) \rangle \langle D_t \left(\frac{\partial f^j}{\partial x_i} \right), h(t) \rangle$$

$$\begin{aligned}
&= \sum_{j=1}^n \sum_{i=1}^n \left\langle h_i^j(s) \left\langle D_t \left(\frac{\partial f^j}{\partial x_i} \right), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle D_t \left(\sum_{i=1}^n h_i^j(s) \frac{\partial f^j}{\partial x_i} \right), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle D_t (DF_j), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle D_h (DF_j), h_j(s) \right\rangle
\end{aligned}$$

and if I haven't made a mistake, then what we have shown so far is that

$$\begin{aligned}
D_h(\delta(u)) &= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \\
&= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n \left\langle D_h (DF_j), h_j \right\rangle
\end{aligned} \tag{3.4} \quad \boxed{\text{P:3.2.4-3}}$$

Also, the above is exactly line two of the first centered equations on page 53.

Now we need to show that the last two terms add up to $\delta(D_h(u))$. This is not defined in the text but I think that we need to define $D_h u$ as the following

$$D_h u = D_h \sum_{j=1}^n F_j h_j(t) = \sum_{j=1}^n h_j(t) D_h(F_j) \tag{3.5} \quad \boxed{\text{P:3.2.4-4}}$$

Thus now $D_h(u)$ (3.5) is of the form $\sum F_j h_j$ and so we may apply the definition of δ to it to get the following

$$\delta(D_h u) = \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle \tag{3.6} \quad \boxed{\text{P:3.2.4-5}}$$

Now we focus in the last term of the right hand side of (3.6). From (3.4) we note that we want to show that

$$\sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle = \sum_{j=1}^n \left\langle D_h (DF_j), h_j \right\rangle \tag{3.7} \quad \boxed{\text{P:3.2.4-7}}$$

Here are the steps,

$$\begin{aligned}
\sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle &= \sum_{j=1}^n \left\langle D_s \left(\left\langle \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(t), h(t) \right\rangle \right), h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle D_s \left(\sum_{i=1}^n \frac{\partial f^j}{\partial x_i} \left\langle h_i^j(t), h(t) \right\rangle \right), h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} \right) \left\langle h_i^j(t), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} \right) h_i^j(t), h(t) \right\rangle, h_j(s) \right\rangle
\end{aligned}$$

$$\begin{aligned}
&= \sum_{j=1}^n \left\langle \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} h_i^j(t) \right), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle D_s(D_t F_j), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle D_h(D F_j), h_j \right\rangle
\end{aligned}$$

Hence (3.7) is shown and we are done.

Proof of bottom of page 52

$$\begin{aligned}
\sum_{i,j=1}^{\infty} D_{e_i} \langle u, e_j \rangle D_{e_j} \langle v, e_i \rangle &= \sum_{i,j=1}^{\infty} \left\langle D_s \langle u, e_j \rangle, e_i(s) \right\rangle \left\langle D_t \langle v, e_i \rangle, e_j(t) \right\rangle \\
&= \sum_{i,j=1}^{\infty} \left\langle \langle D_s(u_t), e_j(t) \rangle, e_i(s) \right\rangle \left\langle \langle D_t(v_s), e_i(s) \rangle, e_j(t) \right\rangle \\
&= \sum_{i=1}^{\infty} \left\langle \sum_{j=1}^{\infty} \left(\langle D_s(u_t), e_j(t) \rangle \left\langle \langle D_t(v_s), e_i(s) \rangle, e_j(t) \right\rangle \right), e_i(s) \right\rangle \\
&= \sum_{i=1}^{\infty} \left\langle \left\langle D_s(u_t), \langle D_t(v_s), e_i(s) \rangle \right\rangle, e_i(s) \right\rangle \\
&= \sum_{i=1}^{\infty} \left\langle \left\langle D_s(u_t), \langle D_t(v_q), e_i(q) \rangle \right\rangle, e_i(s) \right\rangle \\
&= \sum_{i=1}^{\infty} \int_0^{\infty} \left(\int_0^{\infty} D_s(u_t) \left(\int_0^{\infty} D_t(v_q) e_i(q) dq \right) dt \right) e_i(s) ds \\
&= \int_0^{\infty} \left(\int_0^{\infty} D_s(u_t) \left(\sum_{i=1}^{\infty} \int_0^{\infty} D_t(v_q) e_i(q) dq e_i(s) \right) dt \right) ds \\
&= \int_0^{\infty} \int_0^{\infty} D_s(u_t) D_t(v_s) dt ds
\end{aligned}$$

Where above we use the identity

$$\langle f, g \rangle = \sum_{i=1}^{\infty} \langle f, e_i \rangle \langle g, e_i \rangle.$$

Proof of step from property 3

$$\begin{aligned}
E \left(\sum_{i=1}^{\infty} \langle v, e_i \rangle \left(\langle u, e_i \rangle + \delta(D_{e_i}(u)) \right) \right) &= E(\langle u, v \rangle) + E \left(\sum_{i=1}^{\infty} \langle v, e_i \rangle \delta(D_{e_i}(u)) \right) \\
&= E(\langle u, v \rangle) + \sum_{i=1}^{\infty} E \left(\left\langle D(\langle v, e_i \rangle), D_{e_i}(u) \right\rangle \right) \quad (3.8) \quad \boxed{\mathbf{s-1}} \\
&= E(\langle u, v \rangle) + \sum_{i=1}^{\infty} E \left(\left\langle \langle D_s(v_t), e_i(t) \rangle, \langle D_t(u_s), e_i(t) \rangle \right\rangle \right) \\
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(\left\langle \langle D_s(v_t), e_i(t) \rangle, e_j(s) \right\rangle \left\langle \langle D_t(u_s), e_i(t) \rangle, e_j(s) \right\rangle \right)
\end{aligned}$$

$$\begin{aligned}
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(D_{e_j} \langle v_t, e_i(t) \rangle \left\langle \langle D_t(u_s), e_j(s) \rangle, e_i(t) \right\rangle \right) \\
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(D_{e_j} \langle v_t, e_i(t) \rangle D_{e_i} \langle u_s, e_j(s) \rangle \right) \\
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(D_{e_j} \langle v, e_i \rangle D_{e_i} \langle u, e_j \rangle \right)
\end{aligned} \tag{3.9} \quad \boxed{\text{s-2}}$$

where in (3.8) we used Proposition 3.2.3 and in (3.9) we use Fubini's Theorem.

Nick's Question 3.4 (Why is δ linear?) On page 55 it says that δ is linear. First, for $u = \sum_{j=1}^n F_j h_j$ we have

$$\delta(u) = \sum_{j=1}^n F_j B(h_j) - \sum_{j=1}^n \langle DF_j, h_j \rangle_H$$

Now if we have a $v = \sum_{j=1}^n G_j h'_j$ how does $\delta(u+v) = \delta(u) + \delta(v)$. Also how could we write $\sum_{j=1}^n F_j h_j + \sum_{j=1}^n G_j h'_j = \sum_{j=1}^n K_j k_j$ where $K_j \in \mathcal{S}$ and $k_j \in H$

Do we do this through the relationship $E(\langle DF, u \rangle) = E(\delta(u)F)$ in the sense that

$$\begin{aligned}
E(\delta(u+v)F) &= E(\langle DF, u+v \rangle) \\
&= E(\langle DF, u \rangle) + E(\langle DF, v \rangle) \\
&= E(\delta(u)F) + E(\delta(v)F) \\
&= E((\delta(u) + \delta(v))F)
\end{aligned}$$

and if this is true for every F then I believe we can say $\delta(u+v) = \delta(u) + \delta(v)$? **Yes, this is the argument to show that δ is linear. In general, for two distributions F and G , how one identifies $F = G$? This is true iff $\langle F, \varphi \rangle = \langle G, \varphi \rangle$ for all test function φ . Here, is the same. In order to identify that two random variables $\delta(u+v)$ and $\delta(v) + \delta(u)$ are same, this relation has to be tested against all functions in its dual space, which is $\mathbb{D}^{1,2}$.**

Nick's Question 3.5 Exercise 3.1 I don't see how we can apply this hint? Suppose we are able to show that the hint is true then how does that show $\mathcal{P}$ is dense in $L^q(\Omega)$? There is a theorem that $\{x_n\}$ is a basis for a Hilbert space H if and only if $\sum \langle x, x_n \rangle x_n = 0$ implies $x = 0$ but this is only valid for Hilbert space and $L^q(\Omega)$ for $q \neq 2$ is not a Hilbert space since there is no inner product. **This is the standard argument to show one set is dense in another. You may miss the fact that the set of polynomial functions is dense in $C_0^\infty(\mathbb{R}^d)$, the latter is dense in the set of the Schwartz test function $\mathcal{S}(\mathbb{R}^d)$. Please do justify yourself this problem.**

Nick's Question 3.6 Prop 3.3.2 page 54 This relates to exercise 3. In the proof of prop 3.3.2 how can we use Holders inequality to show that $\varphi'(F)D(F) \in L^q(\Omega, H)$?

Nicks Solution.

$$[E \|\varphi'(F)DF\|_H^q]^{1/q} = \left[\int_{\Omega} \left(\int_0^\infty \varphi'(F)^2 (D_t F)^2 dt \right)^{q/2} dP \right]^{1/q}$$

$$\begin{aligned}
&\leq \left[\int_{\Omega} (1 + |F|^{\alpha})^q \left(\int_0^{\infty} (D_t F)^2 dt \right)^{q/2} dP \right]^{1/q} \\
&= \left\| (1 + |F|^{\alpha}) \left(\int_0^{\infty} (D_t F)^2 dt \right)^{1/2} \right\|_q
\end{aligned}$$

Note that $\frac{1}{q} = \frac{1}{p} + \frac{1}{p/\alpha}$ and so if we apply Holders Inequality above we get that

$$[E \|\varphi'(F) DF\|_H^q]^{1/q} \leq \|1 + |F|^{\alpha}\|_{p/\alpha} \left\| \left(\int_0^{\infty} (D_t F)^2 dt \right)^{1/2} \right\|_p$$

which is finite since $F \in \mathbb{D}^{1,p}$. Or equivalently,

$$\begin{aligned}
\|\varphi'(F) DF\|_H^2 &= \|\varphi(F)\|_q^2 \|DF\|_H^2 \\
&\leq \|\varphi(F)\|_{\frac{p}{\alpha}}^2 \|DF\|_p^2 \\
&= \|\varphi(F)\|_{\frac{p}{\alpha}}^2 \int_0^{\infty} \|D_t F\|_p^2 dt
\end{aligned}$$

□

Nick's Question 3.7 Prop 3.3.2 page 54

I have not been able to verify why $\varphi(F) \in \mathbb{D}^{1,q}$. By definition, this means that we need to show that there exists $X_n \in \mathcal{S}$ such that

1. $X_n \rightarrow \varphi(F)$ in $L^q(\Omega)$
2. $D(X_n) \rightarrow D(\varphi(F))$ in $L^q(\Omega, H)$

You have made the problem more complex than what it should be. You only need to check the seminorm $\|\cdot\|_{1,p}$ is finite.

By assumption we already have that there exists $\{F_n\} \subset \mathcal{S}$ such that $F_n \rightarrow F$ in $L^p(\Omega)$ and also $DF_n \rightarrow DF$ in $L^p(\Omega, H)$. We then take a sequence of mollifiers, $\{\alpha_n\}$ and consider the functions $\varphi_k(x) = (\varphi * \alpha_k)(x)$ which by the original assumption on φ are $\alpha + 1$ holder continuous. You cannot say a function is β -Hölder continuous with $\beta \geq 1$. That means the function is a constant.

Now, we notice that by properties of the mollifier since $\varphi(F) \in L^q(\Omega)$ then

$$E|\varphi_k(F_j) - \varphi(F_j)|^q \xrightarrow{j \rightarrow \infty} 0$$

Next, since φ is $\alpha + 1$ holder continuous then we have

$$\begin{aligned}
E|\varphi(F_j) - \varphi(F)|^q &\leq KE|F_j - F|^{q(\alpha+1)} \\
&= KE|F_j - F|^p \rightarrow 0
\end{aligned}$$

and thus we conclude as k and j go to infinity that $\varphi_k(F_j) \rightarrow \varphi(F)$ in $L^q(\Omega)$ and $\varphi_k(F_j) \in \mathcal{S}$.

I am unable to show why $D_t(\varphi_k(F_j)) \rightarrow D_t(\varphi(F))$ in $L^q(\Omega, H)$.

However since $F_j = f^j(B(h_1^j), \dots, B(h_n^j))$, we see that

$$D_t(\varphi_k(F_j)) = \sum_{i=1}^n \frac{\partial}{\partial x_i} (\varphi * \alpha_k)(F_j) h_i^j(t)$$

$$\begin{aligned}
&= \sum_{i=1}^n (\varphi' * \alpha_k)(F_j) \frac{\partial f^j}{\partial x_i} h_i^j(t) \\
&= (\varphi' * \alpha_k)(F_j) \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(t) \\
&= (\varphi' * \alpha_k)(F_j) D_t(F_j).
\end{aligned}$$

We know that $\{\alpha_k\}$ are mollifiers so $(\varphi' * \alpha_k)(F_j) \rightarrow \varphi'(F_j)$ almost surely and by assumption $D_t(F_j) \rightarrow D_t(F)$ in $L^p(\Omega, H)$. But I still am having trouble rigorously showing that $D_t(\varphi_k(F_j)) \rightarrow D_t(\varphi(F_j))$ in $L^p(\Omega, H)$. **You have two indices, you need to take the limit once for just one indices, not the two at the same time. More precisely,**

$$\|D_t(\varphi_k(F_j) - D_t(\varphi(F)))\|_p \leq \|D_t(\varphi_k(F_j)) - D_t(\varphi(F_j))\|_p + \|D_t(\varphi(F_j)) - D_t(\varphi(F))\|_p.$$

Actually, you need to split to three such differences. I will explain to you during the meeting.

Nick's Question 3.8 Exercise 3.4 Let $M = \sum_{t \in [0,1]} X_t$. The hint says to approximate this by taking the sup over a finite set. We know that since X_t is continuous in t then $M = \sup_{t \in \mathbb{A}} X_t$ where $A = \mathbb{Q} \cap [0, 1]$. Let $\{a_n\}$ be an enumeration of $A \setminus \{0, 1\}$ and let $S_k = \{0, a_1, \dots, a_k, 1\}$ then each S_k are finite and $M_k := \sup_{t \in S_k} X_t$ is an **(Note how to type $\mathbb{Q}$, you need curly braces.)** approximation of M since $M_k \rightarrow M$ almost surely. The hint also tells us to use Corollary 4.2.5 but that requires us to find $D_t M_k$ which I am unsure of how to do. The hint says that we need to use exercise 3.3 which is chain rule for Lipschitz functions but I do not know how that applies. This corollary would also require us to show that $M_k \xrightarrow{L^2(\Omega)} M$ which I have been unable to show either. **Try harder... This is an interesting problem that is related the hitting problems that Robert Dalang and Eulalia have done a lot work in the SPDE setting.**

3.2 Exercises

Exercise 3.3 Let α_n be a sequence of mollifiers and $\varphi_k := (\varphi * \alpha_k)$. Since φ is Lipschitz with Lipschitz constant K then this implies that $\varphi'_k \leq K$ and in addition φ' exists almost everywhere and is also bounded by K . By proposition 3.3.2 we have that

$$D_t(\varphi_k(F)) = \varphi'_k(F) D_t(F)$$

Also note that

$$\begin{aligned}
|\varphi_k(F) - \varphi(F)|^2 &= \left| \int_{\mathbb{R}} (\varphi(F - y) - \varphi(F)) \alpha_k(y) dy \right|^2 \\
&\leq K \left| \int_{\mathbb{R}} y \alpha_k(y) dy \right|^2 \\
&\leq K'
\end{aligned}$$

Hence $|\varphi_k(F) - \varphi(F)|^2$ is bounded above and also converges to 0 almost surely so by the dominated convergence theorem we have that

$$\mathbb{E}|\varphi_k(F) - \varphi(F)|^2 \rightarrow 0$$

and so the hypothesis of corollary 4.2.5 is satisfied. This implies that $\varphi(F) \in \mathbb{D}^{1,2}$ and

$$D_t(\varphi_n(F)) \rightarrow D_t(\varphi(F))$$

weakly in $L^2(\Omega, H)$. Lastly, since $\varphi'_k(F)D_t(F)$ is bounded in $L^2(\Omega, H)$ and also

$$\varphi'_k(F)D_t(F) \rightarrow GD_t(F)$$

almost surely where G is a random variable bounded by K then

$$\varphi'_k(F)D_t(F) \rightarrow GD_t(F)$$

weakly in $L^2(\Omega, H)$ as well. Since weak limits are unique then we must have

$$D_t(\varphi(F)) = GD_t(F)$$

However note that in general we have as a property of mollifiers that $\varphi'_n(x) \rightarrow \varphi'(x)$ almost surely. But we said before that $\varphi'(x)$ only exists almost surely since φ is Lipschitz. Therefore if the law of F is absolutely continuous with respect to Lebesgue measure then we have $G = \varphi'(F)$ almost surely. Indeed, let $\mathcal{N}$ be the null set where φ' is undefined. If the law of F is absolutely continuous with respect to Lebesgue measure then $P(F \in \mathcal{N}) = 0$ and we must have $G = \varphi'(F)$ almost surely.

Exercise 3.4 I found a variant of this problem [here](#) where they consider $\{B_t\}_{t \in [0,1]}$ instead of an arbitrary centered Gaussian processes $\{X_t; t \in [0,1]\}$. They use the idea I was trying, which is we want to consider the function $f(x) = \max_{i \in \{1, \dots, n\}} x_i$ for $x \in \mathbb{R}^n$. f is a Lipschitz function and if we consider $\{q_n\}$ an enumeration of $\mathbb{Q} \cap [0,1]$ and consider $Y_n = (X_{q_1}, \dots, X_{q_n})$ then $M_n := \max_{i \in \{1, \dots, n\}} \{X_{q_1}, \dots, X_{q_n}\} = f(Y_n)$ and $M_n \rightarrow M$ almost surely. However in the above linked pdf they claim without proof that that $M_n \rightarrow M$ in $L^2(\Omega)$ which I have so far been unable to show.

I was unaware that there is an extension of the chain rule for Lipschitz functions that is given in the above linked pdf that says we may consider Lipschitz functions on $\mathbb{R}^d$ and random vectors $F = (F_1, \dots, F_n)$ whose components are in $\mathbb{D}^{1,2}$ and then $f(F)$ will also be in $\mathbb{D}^{1,2}$. We apply this above to see that $M_n := \max_{i \in \{1, \dots, n\}} \{X_{q_1}, \dots, X_{q_n}\} = f(Y_n) \in \mathbb{D}^{1,2}$. Next we define the sets

$$A_1 = \{M_n = X_{q_1}\}$$

...

$$A_k = \{M_n \neq X_{q_1}, M_n \neq X_{q_2}, \dots, M_n = X_{q_k}\}, 2 \leq k \leq n$$

and by doing so

$$\begin{aligned} DM_n &= D \left(\sum_{k=1}^n 1_{A_k} X_{q_k} \right) \\ &= \sum_{k=1}^n 1_{A_k} D(X_{q_k}) \end{aligned}$$

where the last equality is to follow because Lemma 3.4.2 on page 56 but I need to verify why. Lets suppose for now that $\{X_t\}$ are Brownian motions. Then

$$D(X_{q_k}) = 1_{[0, q_k]}(t) \tag{3.10} \quad \boxed{\text{brown}}$$

and the above becomes

$$\begin{aligned} DM_n &= \sum_{k=1}^n 1_{A_k} D(X_{q_k}) \\ &= \sum_{k=1}^n 1_{A_k} 1_{[0, q_k]}(t) \end{aligned}$$

and we have that

$$\begin{aligned}\mathbb{E} \|DM_n\|_H^2 &= \mathbb{E} \left\| \sum_{k=1}^n 1_{A_k} 1_{[0,q_k]}(t) \right\|_H^2 \\ &= \mathbb{E} \int \left(\sum_{k=1}^n 1_{A_k} 1_{[0,q_k]}(t) \right)^2 dt \\ &= \sum_{k=1}^n \mathbb{E} \left(1_{A_k} \int 1_{[0,q_k]}(t) dt \right)\end{aligned}$$

where the last inequality follows from the disjointness of the A_k . Continuing the calculations we see

$$\begin{aligned}\mathbb{E} \|DM_n\|_H^2 &= \sum_{k=1}^n \mathbb{E} \left(1_{A_k} \int 1_{[0,q_k]}(t) dt \right) \\ &= \sum_{k=1}^n q_k \mathbb{E} (1_{A_k}) \\ &\leq \sum_{k=1}^n \mathbb{E} (1_{A_k}) \\ &\leq \mathbb{E} (\Omega) = 1\end{aligned}$$

and so we apply Corollary 4.2.5 to conclude that $M = \sup_{t \in [0,1]} X_t \in \mathbb{D}^{1,2}$ and that $DM_n \rightarrow DM$ in the weak topology of $L^2(\Omega, H)$ which means that for all $G \in \mathbb{D}^{1,2}$

$$\langle DM_n, G \rangle_{L^2(\Omega, H)} \rightarrow \langle DM, G \rangle_{L^2(\Omega, H)}$$

To prove that $DX = 1_{[0,T]}$ we note that $\sum_{k=1}^n 1_{A_k} q_k$ which converges almost surely to T . Then we can conclude that $DM_n = \sum_{k=1}^n 1_{A_k} 1_{[0,q_k]}(t)$ converges to $1_{[0,T]}$ almost surely. Since DM_n converges both weakly and almost surely then a result from functional analysis says that these limits must be equal and we see that $DM = 1_{[0,T]}$.

Now if we go back to (3.10) and assume that $\{X_t\}$ is a centered Gaussian process then I am unsure of how to continue. For example we can not calculate $D(X_t)$ for any $t \in [0, 1]$ unless we know that $X_t = \int_0^\infty h(s) dB_s$ for some $h \in H$ but I do not think that this can be claimed.

4 Chapter 4

4.1 Reading material

Nick's Question 4.1 proof of proposition 4.2.1 In the conclusion of this proof, Nualart refers to a density argument but what is he referring too? We see that from (4.1) on page 64 that

$$\mathbb{E} I_n^2(f) = n! \left\| \tilde{f} \right\|_{L^2(\mathbb{R}_+^n)}$$

so is there a result that says that functions

$$A := \{g^{\otimes n}(x_1, \dots, x_n) = g(x_1) \cdots g(x_n), g \in L^2(\mathbb{R}^+)\}$$

are dense in $L_s^2(\mathbb{R}_+^n)$? If that's the case then I believe that for a given $f \in L_s^2(\mathbb{R}_+^n)$ then we

find $\{g_k\} \in A$ such that

$$\mathbb{E}|I_n(f - g_k)|^2 = n! \left\| \tilde{f} - g_k \right\|_{L^2(\mathbb{R}_+^n)}^2 \rightarrow 0$$

and I think since D is a closed operator then we need

$$D_t(I_n(g_k)) = nI_{n-1}(g_k(\cdot, t)) \rightarrow D_t(I_n(f))?$$

Nick's Question 4.2 Example 4.2.3 We know that $B_1^3 = \sum_{n=0}^{\infty} I_n(f_n)$ and by the bottom of page 66, for this example we use the formula

$$f_n = \frac{1}{n!} \mathbb{E}(D^n B_1^3)$$

and calculating each of the above we see that

$$\begin{aligned} f_0 &= \mathbb{E}B_1^3 = 0 \\ f_1 &= 31_{[0,1]}(t_1) \\ f_2 &= 0 \\ f_3 &= 1_{[0,1]}(t_1)1_{[0,1]}(t_2)1_{[0,1]}(t_3) \\ f_k &= 0, \quad k \geq 4 \end{aligned}$$

thus we see that by using the definition on page 63 that

$$\begin{aligned} B_1^3 &= I_1(31_{[0,1]}(t_1)) + I_3(1_{[0,1]}(t_1)1_{[0,1]}(t_2)1_{[0,1]}(t_3)) \\ &= \int_0^\infty 31_{[0,1]}(t_1)dB_{t_1} + \int_0^\infty \int_0^{t_3} \int_0^{t_2} 1_{[0,1]}(t_1)1_{[0,1]}(t_2)1_{[0,1]}(t_3)dB_{t_1}dB_{t_2}dB_{t_3} \\ &= 3B_1 + \int_0^1 \int_0^{t_3} 1_{[0,1]}(t_2) \int_0^{t_2} 1_{[0,1]}(t_1)dB_{t_1}dB_{t_2}dB_{t_3} \\ &= 3B_1 + \int_0^1 \int_0^{t_3 \wedge 1} \int_0^{t_2 \wedge 1} dB_{t_1}dB_{t_2}dB_{t_3} \\ &= 3B_1 + \int_0^1 \int_0^{t_3} \int_0^{t_2} dB_{t_1}dB_{t_2}dB_{t_3} \end{aligned}$$

This is different from the answer in the test on page 67. Is mine correct?

Nick's Question 4.3 directional derivative page 69

1. $\tau_g(\omega) = \omega + g$ where g is continuous on $\mathbb{R}^+$ and $g(0) = 0$. Why does this imply that

$$\tau_g(B(h)) = B(h) + \langle h, \dot{g} \rangle_H.$$

when by definition

$$\tau_g(B(h)) = B(h) + g$$

but also $B(h) \notin \Omega$ where Ω is the Wiener space and

$$B(h) = \int_0^\infty h(s)dB_s$$

so why does $\tau_g(B(h))$ exist?

2. Why does Girsanov's theorem imply that if $F \in L^2(\Omega)$ then $F(\omega + g)$ is almost surely defined?
3. What is the motivation behind the definition of the directional derivative

$$\tilde{D}_g F = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (F \circ \tau_{\epsilon g} - F)?$$

In this pdf they claim that when we have $DF = \sum_{i=1}^n \partial_i f(W(h_1), \dots, W(h_n)) h_i$ then the directional derivative is

$$\langle DF, h \rangle_H = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (f(W(h_1) + \epsilon \langle h_1, h \rangle, \dots, W(h_n) - \epsilon \langle h_n, h \rangle) - F) \quad (4.1) \quad \text{pdf 1}$$

and also it claims that $\langle DF, h \rangle_H$ is the derivative at $\epsilon = 0$ of the shifted processes

$$F(W(g) + \epsilon \langle g, h \rangle) \quad (4.2) \quad \text{pdf 2}$$

for $g \in H$ but what does this mean?

Consider $F = (W(h_1), \dots, W(h_n))$ and here we have

$$DF = \sum_{i=1}^n \frac{\partial f}{\partial x_i} (W(h_1), \dots, W(h_n)) h_i$$

and so if we try to calculate $\langle DF, h \rangle$ we get

$$\begin{aligned} \langle DF, h \rangle &= \sum_{i=1}^n \frac{\partial f}{\partial x_i} (W(h_1), \dots, W(h_i), \dots, W(h_n)) \langle h_i, h \rangle \\ &= \sum_{i=1}^n \lim_{\epsilon \rightarrow 0} \frac{f(W(h_1), \dots, W(h_i) + \epsilon \langle h_i, h \rangle, \dots, W(h_n)) - F}{\epsilon \langle h_i, h \rangle} \langle h_i, h \rangle \\ &= \sum_{i=1}^n df \lim_{\epsilon \rightarrow 0} \frac{f(W(h_1), \dots, W(h_i) + \epsilon \langle h_i, h \rangle, \dots, W(h_n)) - F}{\epsilon} \end{aligned} \quad (4.3) \quad \text{ddiv}$$

but I do not see how we can simplify this to (4.1)?

Solution for 3 (4.1): The definition of the directional derivative is

$$D_v(f) = \lim_{h \rightarrow 0} \frac{f(x + hv) - f(x)}{h} = \frac{d}{d\epsilon} f(x + \epsilon v)|_{\epsilon=0}$$

and according to wikipedia, when the partials of f exist then it can be shown that

$$D_v(f) = \sum_{i=1}^n v_i \frac{\partial f}{\partial x_i} \quad (4.4) \quad \text{dirdiv}$$

where $v = (v_1, \dots, v_n)$. This is stated without proof and is supposed to follow from the definition of the total derivative which is something I am not familiar too. However, if we let $v = (\langle h_1, h \rangle, \dots, \langle h_n, h \rangle)$ then

$$\begin{aligned} D_v(f) &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (f(W(h_1) + \epsilon \langle h_1, h \rangle, \dots, W(h_n) - \epsilon \langle h_n, h \rangle) - F) \\ &= \sum_{i=1}^n \langle h_i, h \rangle \frac{\partial f}{\partial x_i} \end{aligned}$$

$$= \langle DF, h \rangle$$

where the second equality follows from (4.4) and the third equality follows from the work listed in the gray text from the question I asked.

Nick's Question 4.4 Proof of Theorem 4.4.2

1. Why does

$$\mathbb{E} \left(\left| \frac{1}{\epsilon} (F_n \circ \tau_{\epsilon g} - F_n) - \langle DF_n, \dot{g} \rangle_H \right| \right) \rightarrow 0$$

for fixed n and $\epsilon \rightarrow 0$?

I believe that they are claiming without proof that if $F_n \in \mathcal{S}$ then

$$\frac{1}{\epsilon} (F_n \circ \tau_{\epsilon g} - F_n) \rightarrow \langle DF_n, \dot{g} \rangle_H$$

I assume in $L^2(\Omega)$ as $\epsilon \rightarrow 0$.

I believe this is true in the most simple example where $F_n = \int_0^T h_n(s) dB_s$ where $h_n \in H$ and B_s is the canonical brownian motion on the Wiener space, ie $B_s(\omega) = \omega(s)$. For any $g \in \mathcal{H}$, or in other words

$$g(t) = \int_0^t \dot{g}(s) ds$$

then we can easily calculate the following derivative

$$\begin{aligned} \tilde{D}_g(F_n)(\omega) &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (F_n \circ \tau_{\epsilon g} - F_n) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left(\int_0^t h_n(s) dB_s(\omega + \epsilon g) - \int_0^t h_n(s) dB_s(\omega) \right) \\ &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left(\int_0^t h_n(s) d(\omega(s) + \epsilon g(s)) - \int_0^t h_n(s) d(\omega(s)) \right) \\ &= \lim_{\epsilon \rightarrow 0} \left(\int_0^t h_n(s) dg(s) \right) \\ &= \int_0^t h_n(s) \dot{g}(s) ds \end{aligned}$$

and from this we conclude that

$$\tilde{D}_g(F_n)(\omega) = \langle DF_n, \dot{g} \rangle_H.$$

Can we extend this to all $F_n \in \mathcal{S}$?

2. Why does

$$\frac{1}{\epsilon} (F_n \circ \tau_{\epsilon g} - F_n) = \frac{1}{\epsilon} \int_0^\epsilon \langle DF_n \circ \tau_{\alpha g}, \dot{g} \rangle_H d\alpha$$

when $F_n \in \mathcal{S}$. Above I showed that for $F = \int_0^T h(s) dB_s$ that we have

$$\frac{1}{\epsilon} (F_n \circ \tau_{\epsilon g} - F_n) = \langle h, \dot{g} \rangle_H$$

Also since

$$DF \circ \tau_{\epsilon g} = h(t) \circ \tau_{\epsilon g} = h(t)$$

because in this case DF is constant in ω then we see that

$$\langle h, \dot{g} \rangle_H = \frac{1}{\epsilon} \int_0^\epsilon \langle DF_n \circ \tau_{\alpha g}, \dot{g} \rangle_H d\alpha$$

so the statement is true for the most simple $F \in \mathcal{S}$. However I do not know how to show this for arbitrary $F \in \mathcal{S}$.

3. I don't see how Girsanov's theorem is used to conclude the convergence on the top of page 71? Also how is Girsanov's theorem used in steps 3 and 4 on page 72?

solution to 1:

The question is whether or not $F \in \mathcal{S}$ implies that the directional derivative exists and whether or not

$$\tilde{D}_g(F) = \langle DF, \dot{g} \rangle_H, \quad \text{where } g = \int \dot{g}$$

I showed above in the gray text that this is true when F takes on the most simply form of

$$F = \int_0^T h(s) dB_s$$

where $h \in H$ and B_s is the canonical brownian motion on the Weiner space. However we want this to be true for any general $F \in \mathcal{S}$ or in other words $F = f(B(h_1), \dots, B(h_n))$ where $f \in C_p^\infty(\mathbb{R}^n)$. However, such f can be uniformly approximated by functions of the form x^n so I believe that if we can show that for smooth random variables

$$F = \left(\int_0^T h(s) dB_s \right)^n$$

that

$$\tilde{D}_g(F) = \langle DF, \dot{g} \rangle$$

then we are done. However this is easily shown as follows

$$\begin{aligned} \frac{F \circ \tau_{\epsilon g} - F}{\epsilon} &= \frac{\left(\int_0^T h(s) dB_s(\omega + \epsilon g) \right)^n - \left(\int_0^T h(s) dB_s(\omega) \right)^n}{\epsilon} \\ &= \frac{\left(\int_0^T h(s) d(\omega(s)) + \epsilon \int_0^T h(s) d(g(s)) \right)^n - \left(\int_0^T h(s) d(\omega(s)) \right)^n}{\epsilon} \\ &= \frac{\sum_{i=1}^n \binom{n}{i} \epsilon^i \left(\int_0^T h(s) d(\omega(s)) \right)^{n-i} \left(\int_0^T h(s) \dot{g}(s) ds \right)^i}{\epsilon} \\ &= n \left(\int_0^T h(s) d(\omega(s)) \right)^{n-1} \langle h, \dot{g} \rangle_H + \sum_{i=2}^n \binom{n}{i} \epsilon^{i-1} \left(\int_0^T h(s) d(\omega(s)) \right)^{n-i} \left(\int_0^T h(s) \dot{g}(s) ds \right)^i \\ &\rightarrow n \left(\int_0^T h(s) d(\omega(s)) \right)^{n-1} \langle h, \dot{g} \rangle_H \quad \text{as } \epsilon \rightarrow 0 \\ &= \langle DF, \dot{g} \rangle_H \end{aligned}$$

which is what we wanted to show.

Partial solution for 2: Suppose that $F = \left(\int_0^\infty h(t) dB_t \right)^2$. Then we can calculate the following

$$F \circ \tau_{\epsilon g} - F = \left(\int_0^\infty h(t) dB_t(\omega + \epsilon g) \right)^2 - \left(\int_0^\infty h(t) dB_t(\omega) \right)^2$$

$$\begin{aligned}
&= 2\epsilon \int_0^\infty h(t)dB_t(\omega)\langle h, \dot{g} \rangle + \epsilon^2 \langle h, \dot{g} \rangle^2 \\
&= \epsilon \langle DF, \dot{g} \rangle + \epsilon^2 \langle h, \dot{g} \rangle^2
\end{aligned} \tag{4.5} \quad \boxed{\text{lhs}}$$

In addition, we can also calculate

$$\begin{aligned}
\int_0^\epsilon \langle DF \circ \tau_{\epsilon g}, \dot{g} \rangle d\alpha &= \int_0^\epsilon \int_0^\infty D_s F \circ \tau_{\epsilon g} \dot{g}(s) ds d\alpha \\
&= \int_0^\epsilon \int_0^\infty \left(2h(s) \int_0^\infty h(t) dB_t(\cdot) \right) \circ \tau_{\epsilon g} \dot{g}(s) ds d\alpha \\
&= \int_0^\epsilon \int_0^\infty \left(2 \int_0^\infty h(t) dB_t(\omega + \epsilon g) \right) h(s) \dot{g}(s) ds d\alpha \\
&= \int_0^\epsilon \int_0^\infty \left(2 \int_0^\infty h(t) dB_t(\omega) + 2\epsilon \int_0^\infty h(t) \dot{g}(t) dt \right) h(s) \dot{g}(s) ds d\alpha \\
&= \int_0^\epsilon \int_0^\infty 2 \int_0^\infty h(t) dB_t(\omega) h(s) \dot{g}(s) ds d\alpha + \int_0^\epsilon \int_0^\infty 2\epsilon \langle h, \dot{g} \rangle h(s) \dot{g}(s) ds d\alpha \\
&= \epsilon \langle DF, \dot{g} \rangle + \epsilon^2 \langle h, \dot{g} \rangle^2
\end{aligned} \tag{4.6} \quad \boxed{\text{rhs}}$$

thus we see that (4.6) and (4.5) imply that

$$\frac{1}{\epsilon} \int_0^\epsilon \langle DF \circ \tau_{\epsilon g}, \dot{g} \rangle d\alpha = \frac{1}{\epsilon} (F \circ \tau_{\epsilon g} - F)$$

and so the claim is true for $F = \left(\int_0^\infty h(t) dB_t \right)^2$ which leads us to believe that it is true for $F = \left(\int_0^\infty h(t) dB_t \right)^n$ which again leads us to believe that it is true for all $F \in \mathcal{S}$.

Informal explanation for some of Step 4: The only line that I can't quite convince myself is the first centered equation with says

$$\frac{1}{\alpha} (E((F \circ \tau_{\alpha g})G) - E(FG)) = E \left(F \frac{1}{\alpha} \left((G \circ \tau_{-\alpha g}) \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) - G \right) \right)$$

I don't understand why we pass the $\tau_{\alpha g}$ from F to G and it then becomes a $\tau_{-\alpha g}$. Also is not formally stated but the first expectation is under Q and the second is under P with $Q(A) = E(1_A L_t)$ and here

$$L_t = \exp \left(\int_0^t \dot{g}_s dB_s - \frac{1}{2} \int_0^t \dot{g}_s^2 ds \right)$$

where $\dot{g} = h$ which is stated on the previous page.

The next three centered equations are

$$\begin{aligned}
&\lim_{\alpha \rightarrow 0} \frac{1}{\alpha} \left((G \circ \tau_{-\alpha g}) \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) - G \right) \\
&= \frac{d}{d\alpha} \left((G \circ \tau_{-\alpha g}) \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) \right) \Big|_{\alpha=0}
\end{aligned} \tag{4.7}$$

$$= B(h)G - \langle DG, h \rangle_H \tag{4.8}$$

$$= \delta(Gh) \tag{4.9}$$

The first equality follows from the limit definition of the derivative. The second equality follows from products rule. We have

$$\begin{aligned}
&\frac{d}{d\alpha} \left((G \circ \tau_{-\alpha g}) \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) - G \right) \Big|_{\alpha=0} \\
&= \left(\frac{d}{d\alpha} (G \circ \tau_{-\alpha g}) \right) \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) \Big|_{\alpha=0} + (G \circ \tau_{-\alpha g}) \left(\frac{d}{d\alpha} \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) \right) \Big|_{\alpha=0}
\end{aligned}$$

Note that since $G \in \mathcal{S}$ then we have that $G = g(B(h_1), \dots, B(h_n))$ and therefore

$$G \circ \tau_{-\alpha g} = g(B(h_1), \dots, B(h_n))(\omega - \alpha g) = g(B(h_1) - \alpha \langle h_1, \dot{g} \rangle, \dots, B(h_n) - \alpha \langle h_n, \dot{g} \rangle)$$

and from this we can calculate $d/d\alpha$

$$\begin{aligned} \left. \frac{d}{d\alpha} (G \circ \tau_{-\alpha g}) \right|_{\alpha=0} &= \lim_{k \rightarrow 0} \frac{1}{k} (g(B(h_1) - (\alpha + k) \langle h_1, \dot{g} \rangle, \dots, B(h_n) - (\alpha + k) \langle h_n, \dot{g} \rangle) - G) \Big|_{\alpha=0} \\ &= -\langle DG, \dot{g} \rangle \end{aligned}$$

where the last inequality follows from the work shown in (4.3) which I believe can be applied because again $G \in \mathcal{S}$. Lastly, we simply take the derivative to see

$$\left(\frac{d}{d\alpha} \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) \right) \Big|_{\alpha=0} = B(h)$$

putting this together give us

$$\begin{aligned} \left. \frac{d}{d\alpha} \left((G \circ \tau_{-\alpha g}) \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) - G \right) \right|_{\alpha=0} &= B(h)G - \langle DF, \dot{g} \rangle_H \\ &= B(h)G - \langle DF, h \rangle_H \end{aligned}$$

which completes the verification for step 4.

Nick's Question 4.5 The formula

$$I_n(f)I_m(g) = \sum_{r=0}^{n \wedge m} r! \binom{n}{r} \binom{m}{r} I_{n+m-2r}(f \otimes_r g)$$

on page 64:

This also relates to exercise 2. I tried testing this formula out for the most basic example, for $f, g \in L^2(\mathbb{R}_+)$ and $I_1(f)I_1(g)$.

Using the formula we would need to get

$$I_1(f)I_1(g) = I_2(f \otimes_0 g) + I_0(f \otimes_1 g) \tag{4.10}$$

$$= \int_0^\infty \int_0^{t_2} f(t_1)g(t_2)dB_{t_1}dB_{t_2} + \int_0^\infty f(x)g(x)dx \tag{4.11}$$

where $(f \otimes_r g)$ is defined on page 64.

But suppose that we do the following:

$$\begin{aligned} X_t &= \int_0^t f(s)dB_s \\ Y_t &= \int_0^t g(s)dB_s \end{aligned}$$

then by applying Ito's formula we see that

$$\begin{aligned} X_t Y_t &= \int_0^t X_s dY_s + \int_0^t Y_s dX_s + \int_0^t dX_s dY_s \\ &= \int_0^t \left(\int_0^s f(\xi)dB_\xi \right) g(s)dB_s + \int_0^t \left(\int_0^s g(\xi)dB_\xi \right) f(s)dB_s + \int_0^t f(s)g(s)ds \\ &= \int_0^t \left(\int_0^s f(\xi)g(s)dB_\xi \right) dB_s + \int_0^t \left(\int_0^s g(\xi)f(s)dB_\xi \right) dB_s + \int_0^t f(s)g(s)ds \end{aligned}$$

eqn

$$\begin{aligned}
&= \int_0^t \int_0^s f(\xi)g(s) + g(\xi)f(s)dB_\xi dB_s + \int_0^t f(s)g(s)ds \\
&= \int_0^t \int_0^s 2(f\tilde{\otimes}_0 g)(\xi, s)dB_\xi dB_s + \int_0^t f(s)g(s)ds
\end{aligned}$$

where above $(f\tilde{\otimes}_0 g)$ is the symmetrization of $(f \otimes_0 g)$. Thus by letting $t \rightarrow \infty$ we see that

$$\begin{aligned}
\lim_{t \rightarrow \infty} X_t Y_t &= \int_0^\infty f(s)dB_s \int_0^\infty g(s)dB_s = I_1(f)I_1(g) \\
&= \int_0^\infty \int_0^s 2(f\tilde{\otimes}_0 g)(\xi, s)dB_\xi dB_s + \int_0^\infty f(s)g(s)ds \\
&= 2I_1(f\tilde{\otimes}_0 g) + I_0(f\otimes_1) \\
&= 2I_1(f \otimes g) + I_0(f\otimes_1) \tag{4.12}
\end{aligned}$$

cont

where the last line follows that $I_n(f) = I_n(\tilde{f})$ for any $f \in L^2(\mathbb{R}_+^n)$. However (4.10) and (4.12) give us a contradiction.

Exercises:

Exercise 2: We want to prove the following for $f, g \in L_s^2(\mathbb{R}_+^2)$

$$I_2(f)I_2(g) = I_4(f \otimes_0 g) + 4I_2(f \otimes_1 g) + 2I_0(f \otimes_2 g).$$

The hint tells us to use Ito's formula. Consider the following,

$$\begin{aligned}
X_t &= \int_0^t \int_0^\xi f(z, \xi)dB_z dB_\xi = \int_0^t F(\xi)dB_\xi \\
Y_t &= \int_0^t \int_0^\xi g(z, \xi)dB_z dB_\xi = \int_0^t G(\xi)dB_\xi
\end{aligned}$$

where we let

$$F(\xi) = \int_0^\xi f(z, \xi)dB_z \quad G(\xi) = \int_0^\xi g(z, \xi)dB_z$$

and so

$$\lim_{t \rightarrow \infty} X_t Y_t = I_2(f)I_2(g).$$

If we apply Ito's formula, we see that

$$X_t Y_t = \int_0^t X_s dY_s + \int_0^t Y_s dX_s + \int_0^t dX_s dY_s := L_1 + L_2 + L_3 \tag{4.13}$$

ito1

If this is correct, then what does dX_s and dY_s equal? For, example, is $dX_s = F(s)dB_s = (\int_0^s f(z, s)dB_z)dB_s$? If this is right, then we can continue with (4.13).

We will just look at L_1 right now.

$$\begin{aligned}
L_1 &= \int_0^t X_s dY_s \\
&= \int_0^t \underbrace{\left(\int_0^s F(\xi)dB_\xi \right)}_{X_s} G(s)dB_s
\end{aligned}$$

notice the integrand is a product of two processes. The only way I think I can continue is to apply Ito's formula again. But the problem gets messy here.

$$X_s G_s = \int_0^s X_k dG_k + \int_0^s G_k dX_k + \int_0^s dX_k dG_k$$

To continue, we need to calculate dG_k which can be done with the following Stochastic Leibniz Rule Theorem which is found [here](#). However I believe we need to assume that g is differentiable in addition to being square integrable. According to Leibniz rule we have that

$$dG_k = \left(\int_0^k \frac{\partial g}{\partial k}(z, k) dB_z \right) dk + g(k, k) dB_k$$

I dont think continuing with this strategy gets us to the desired answer though.

5 Chapter 5:

Nick's Question 5.1 Proposition 5.1.2

1. What does $F(e^{-t}B + \sqrt{1 - e^{-2t}}B')$ mean? What are B and B' ? Why is he dropping the subscripts B_t and B'_t ? I thought this was only notation but it seems that something else is happening as well? It seems like F is being evaluated at $e^{-t}B + \sqrt{1 - e^{-2t}}B'$ but this can't be possible since $e^{-t}B + \sqrt{1 - e^{-2t}}B' \in \mathbb{R}$ and $F \in L^2(\Omega)$. David Nualart [has this pdf](#). Page 5 and page 20 have some info.

2. How does

$$E'(|F(e^{-t}B + \sqrt{1 - e^{-2t}}B')|^p) = E|F|^p?$$

3. Where does

$$E' \left(\exp \left(e^{-t} \lambda B(h) + \sqrt{1 - e^{-2t}} \lambda B'(h) - \frac{1}{2} \lambda^2 \right) \right)$$

come from?

Nick's Question 5.2

Why does Mehler's formula imply that the operator T_t is nonnegative?

Nick's Question 5.3 Theorem 5.1.3

There is a [reference here](#) however it is in French.

1. Why is $|T_t F| \leq |F|$? All I think that we can say is that

$$|T_t F| \leq \sum_n e^{-nt} |I_n(f_n)| \leq \sum_n |I_n(f_n)|$$

and

$$|F| \leq \sum_n |I_n(f_n)|$$

but this does not mean that

$$|T_t(F)| \leq |F|.$$

2. Why does $|T_t F| \leq |F|$ imply that we may assume that F and G are nonnegative?
3. I think there is some typo in the claim that

$$E(\tilde{E}(F(\hat{B}))G) = E(F(\hat{B})G)$$

This is because $E(\tilde{E}(F(\hat{B}))G) \in \mathbb{R}$ while $E(F(\hat{B})G) \in \mathbb{R}^\Omega$. I am not sure what the correction should be.